

Problem 3.23

Model: $\text{PRICE} = \alpha_1 + \alpha_2 \text{SQFT}^2 + e$

Estimated: $\hat{\text{PRICE}} = 93.574 + 0.18452 \cdot \text{SQFT}^2$

$\text{se}(\hat{\alpha}_2) \approx 0.005257$, $N = 500$, $\text{df} = 498$

(a)

Marginal effect of a 100 sq ft increase at $\text{SQFT}_0 = 20$ (2,000 sq ft):

$$\partial E(\text{PRICE}) / \partial \text{SQFT} = 2\alpha_2 \cdot \text{SQFT}_0 = 2\alpha_2 \times 20 = 40\alpha_2$$

Hypotheses:

$$H_0: 40\alpha_2 \leq 13$$

$$H_1: 40\alpha_2 > 13$$

Test statistic:

$$\begin{aligned} t &= (40\hat{\alpha}_2 - 13) / \text{se}(40\hat{\alpha}_2) = (40 \times 0.18452 - 13) / (40 \times 0.005257) \\ &= (7.381 - 13) / 0.2103 = -26.729 \end{aligned}$$

Rejection region: $t > t_{0.05, 498} = 1.648$

p-value = 1.000

Since $t = -26.729 < 1.648$, we fail to reject H_0 at the 5% level. The data do not support the claim that the marginal effect at 2,000 sq ft exceeds \$13,000. The estimated marginal effect is well below \$13,000.

(b)

Marginal effect of a 100 sq ft increase at $\text{SQFT}_0 = 40$ (4,000 sq ft):

$$\partial E(\text{PRICE}) / \partial \text{SQFT} = 2\alpha_2 \cdot \text{SQFT}_0 = 2\alpha_2 \times 40 = 80\alpha_2$$

Hypotheses:

$$H_0: 80\alpha_2 \leq 13$$

$$H_1: 80\alpha_2 > 13$$

Test statistic:

$$\begin{aligned} t &= (80\hat{\alpha}_2 - 13) / \text{se}(80\hat{\alpha}_2) = (80 \times 0.18452 - 13) / (80 \times 0.005257) \\ &= (14.762 - 13) / 0.4206 = 4.189 \end{aligned}$$

Rejection region: $t > t_{0.05, 498} = 1.648$

p-value = $1.654 \times 10^{-5} \approx 0.0000$

Since $t = 4.189 > 1.648$ and p-value < 0.05 , we reject H_0 at the 5% level. The marginal effect of a 100 sq ft increase for a 4,000 sq ft house is significantly greater than \$13,000. The estimated marginal effect is \$14,762.

(c)

For SQFT = 20 (2,000 sq ft):

$$\hat{E}(\text{PRICE} \mid \text{SQFT}=20) = 93.574 + 0.18452 \times (20^2) = 93.574 + 73.808 = 167.374 \text{ (\$167,374)}$$

95% Confidence Interval: (158.048, 176.699) [in \$1,000 units]

We are 95% confident that the expected selling price of a 2,000 sq ft house is between \$158,048 and \$176,699.

(d)

Sample mean price for houses with SQFT = 20: \$163,333 (i.e., 163.333 in \$1,000 units).

The sample mean falls within the 95% CI from part (c): (158.048, 176.699). Therefore, the sample average is compatible with the interval estimate. The point estimate from the regression is close to, but slightly above, the sample mean for houses with exactly 2,000 sq ft. The small difference reflects sampling variation and the fact that the regression-based estimate uses all 500 observations.